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School-Wide Network

Group : 3

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Class	:	2CS1

CEP CCIT

FAKULTAS TEKNIK UNIVERSITAS INDONESIA

2026

PROJECT ON

School-Wide Network

Developed by

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School-Wide Network

Batch Code : 2CS1

Start Date : 19/2/2026

End Date : 3/3/2026

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Date of Submission: 3/3/2026

CERTIFICATE

This is to certify that this report titled “School-Wide Network” embodies the original work done by Aldi Syarif Alhakim, Alezuna M. Nadhif Pohan, Muhamad Ihsan Rasyad. Project in partial fulfillment of their course requirement at CEP-CCIT FTUI.

Coordinator:

M. Riza Iqbal Latief, S.T.

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to God Almighty for the knowledge and strength given to us to complete this project.

We also thank our lecturer, Mr. M. Riza Iqbal Latief, S.T., for his guidance and support throughout this project.

Finally, we appreciate all group members for their cooperation and teamwork in completing the School-Wide Network project.

BACKGROUND

Internet access has become an essential part of the learning process in schools. Students use the internet to access educational materials, complete assignments, conduct research, and support communication activities. However, unmanaged network infrastructure can lead to bandwidth congestion, unstable connections, security vulnerabilities, and inefficient resource distribution.

To address these issues, a School-Wide Network is designed to provide a structured and centralized network infrastructure that connects laboratories, classrooms, and administrative areas. The system integrates wired (LAN) and wireless connections under a single router-based management system to ensure stable and reliable connectivity across the school environment.

By implementing this network design, the school can improve performance, enhance security through network segmentation, manage bandwidth efficiently, and create a more organized and scalable network system to support academic and administrative activities.

SYSTEM ANALYSIS

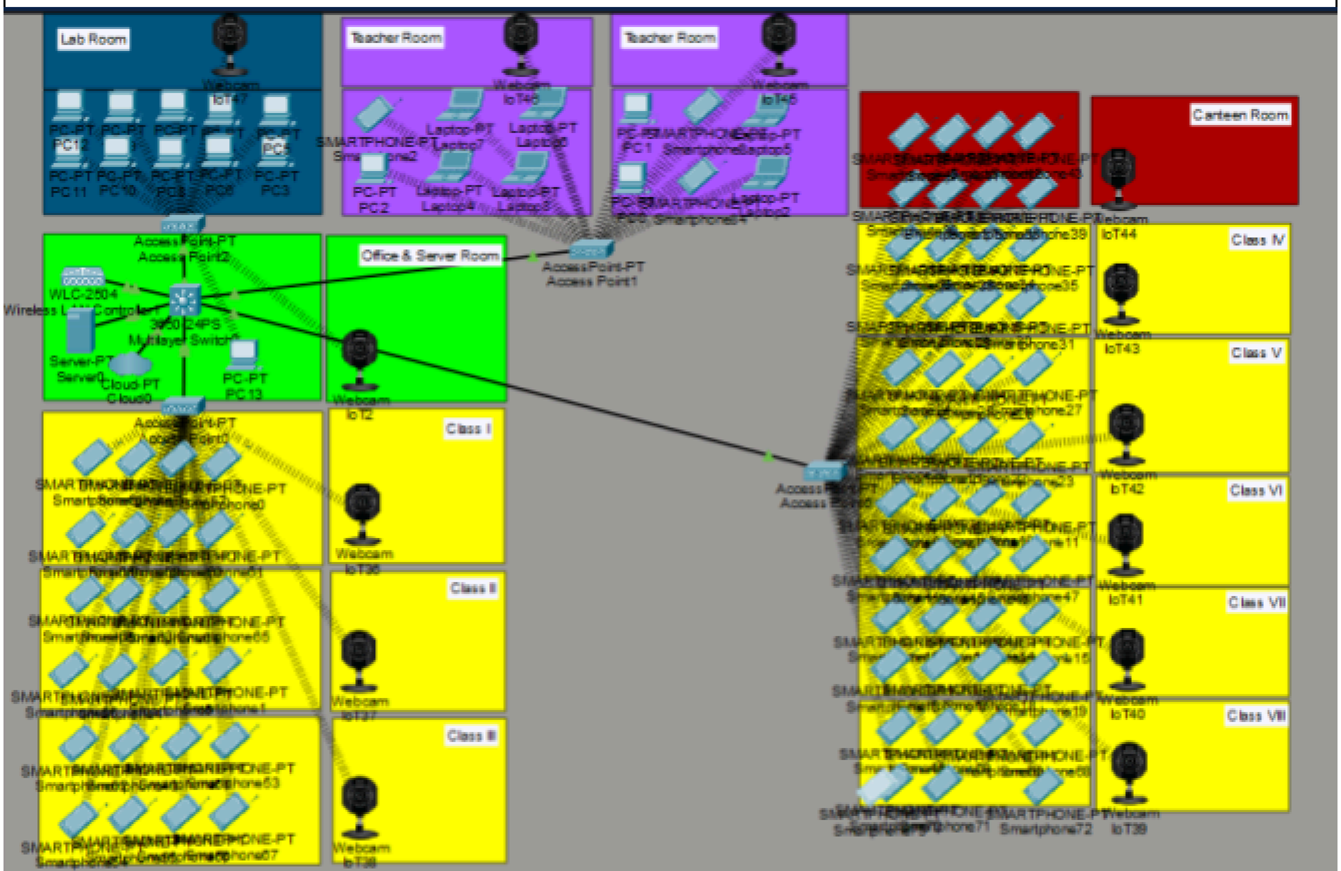
The School-Wide Network is designed to provide centralized and structured network connectivity across the entire school environment. The system supports both wired and wireless connections to ensure stable and efficient communication for academic and administrative activities.

In this design, laboratory computers are directly connected to the server room using wired LAN connections to ensure high stability, low latency, and reliable performance during practical sessions. Meanwhile, classrooms and teacher areas access the network through wireless access points distributed across the school to provide flexible and wide coverage.

The network uses a centralized router connected to an Internet Service Provider (ISP) as the main gateway. The router manages IP addressing, internet distribution, bandwidth control, and network security. Switches are used to distribute connections from the server room to laboratory devices and access points.

By implementing this architecture, the system improves network stability in laboratory environments, provides flexible wireless access for other areas, enhances centralized management, and ensures efficient bandwidth utilization throughout the school.

TOPOLOGY DESIGN



SECTIONS	NETWORK ADDRESS	SUBNETMASK	AVAILABLE HOST
LAB	192.168.40.0	255.255.255.0	253
GURU	192.168.30.0	255.255,255.0	253
SISWA	192.168.20.0	255.255.255.0	253
MURID	192.168.10.0	255.255.255.0	253
MANAGEMENT	192.168.99.0	255.255.255.0	253

NETWORK DEVICES

LAB

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point2	-	=	-
Computer	PC	192.168.40.6 (DHCP) 192.168.40.9 (DHCP) 192.168.40.5 (DHCP) 192.168.40.7 (DHCP) 192.168.40.8 (DHCP) 192.168.40.12 (DHCP) 192.168.40.3 (DHCP) 192.168.40.10 (DHCP) 192.168.40.13 (DHCP) 192.168.40.11 (DHCP)	192.168.40.1	Wireless
Web Cam IOT	Webcam	192.168.40.4 (DHCP)	192.168.40.1	Wireless

TEACHER ROOM

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point1	-	-	-
Computer	PC	192.168.30.12 (DHCP) 192.168.30.2 (DHCP) 192.168.20.11 (DHCP)	192.168.30.1	
Web Cam IOT	Webcam	192.168.30.5 (DHCP) 192.168.30.9 (DHCP)	192.168.30.1	Wireless
SmartPhone	Smartphone	192.168.30.3 (DHCP) 192.168.30.6 (DHCP) 192.168.30.13 (DHCP)	192.168.30.1	
Laptop PT	Laptop	192.168.30.7 (DHCP) 192.168.30.15 (DHCP) 192.168.30.14 (DHCP) 192.168.30.4 (DHCP) 192.168.30.8 (DHCP) 192.168.30.10 (DHCP)	192.168.30.1	

CLASSROOM LEFT

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point6	-	-	-
Smartphone-PT	Smartphone	192.168.10.10 (DHCP) 192.168.10.14 (DHCP) 192.168.10.2 (DHCP) 192.168.10.20 (DHCP) 192.168.10.17 (DHCP) 192.168.10.9 (DHCP) 192.168.10.5 (DHCP) 192.168.10.13 (DHCP) 192.168.10.21 (DHCP) 192.168.10.18 (DHCP) 192.168.10.11 (DHCP) 192.168.10.22 (DHCP) 192.168.10.23 (DHCP) 192.168.10.24 (DHCP) 192.168.10.4 (DHCP) 192.168.10.14 (DHCP) 192.168.10.15 (DHCP) 192.168.10.12 (DHCP) 192.168.10.25 (DHCP) 192.168.10.27 (DHCP) 192.168.10.19 (DHCP) 192.168.10.6 (DHCP)	192.168.10.1	

		192.168.10.7 (DHCP) 192.168.10.3 (DHCP)		
Web Cam IOT	Web Cam	192.168.10.8 (DHCP) 192.168.10.16 (DHCP) 192.168.10.6 (DHCP)	192.168.10.1	WIreless

CANTEEN

Devices	Devices Name	IP Address	Gateway	Features
Web Cam IOT	Webcam	192.168.20.16 (DHCP)	192.168.20.1	Wireless
SmartPhone	Smartphone	192.168.20.20 (DHCP) 192.168.20.14 (DHCP) 192.168.20.36 (DHCP) 192.168.20.19 (DHCP) 192.168.20.22 (DHCP) 192.168.20.24 (DHCP)	192.168.20.1	

CLASSROOM RIGHT

Devices	Devices Name	IP Address	Gateway	Features
Access Point-PT	Access Point6	-	-	-
Smartphone-PT	Smartphone	192.168.20.40 (DHCP) 192.168.20.49 (DHCP) 192.168.20.39 (DHCP) 192.168.20.42 (DHCP)	192.168.20.1	

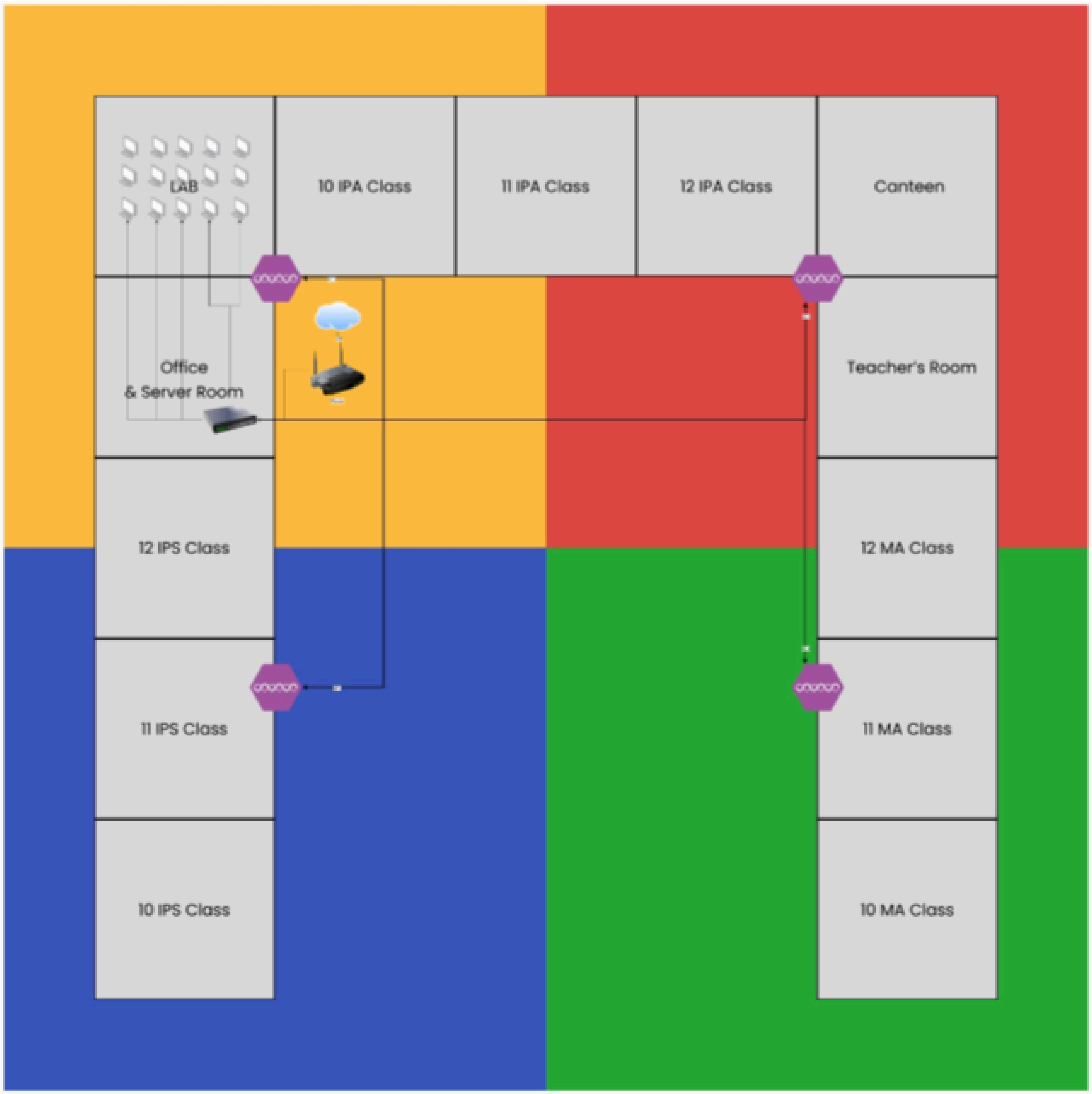
		192.168.20.28 (DHCP) 192.168.20.27 (DHCP) 192.168.20.13 (DHCP) 192.168.20.48 (DHCP) 192.168.20.4 (DHCP) 192.168.20.6 (DHCP) 192.168.20.5 (DHCP) 192.168.20.10 (DHCP) 192.168.20.8 (DHCP) 192.168.20.9 (DHCP) 192.168.20.11 (DHCP) 192.168.20.15 (DHCP) 192.168.20.18 (DHCP) 192.168.20.23 (DHCP) 192.168.20.28 (DHCP) 192.168.20.29 (DHCP) 192.168.20.30 (DHCP) 192.168.20.32 (DHCP) 192.168.20.33 (DHCP) 192.168.20.35 (DHCP)		
WebCam-IOT		192.168.20.27 (DHCP) 192.168.20.34 (DHCP) 192.168.20.25 (DHCP) 192.168.20.31 (DHCP) 192.168.20.58	192.168.20.1 (DHCP)	wireless

		(DHCP)		

Office and Server Room

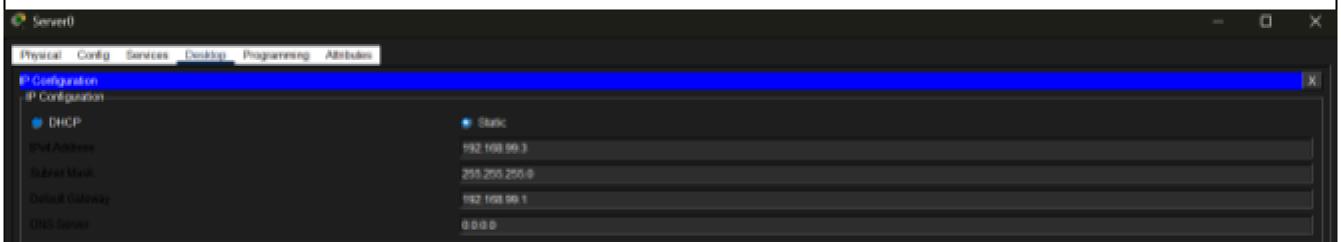
Devices	Devices Name	IP Address	Gateway	Features
Server-PT	server0	192.168.99.3	192.168.99.1	-
Switch	Multilayer Switch	-	-	-
Cloud PT	Cloud	-	-	-
WLC-2504	Wireless LAN C.	192.168.99.2	-	-
PC-PT	Computer	192.168.40.15 (DHCP)	192.168.40.1	
WebCam-IOT	Webcam	192.168.40.2 (DHCP)	192.168.40.1	-Wireless

MAP DESIGN

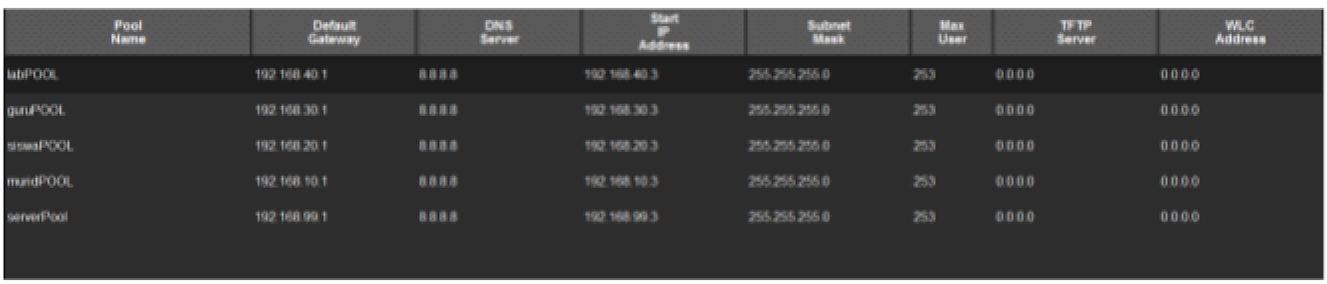

LAB
10 IPA Class
11 IPA Class
12 IPA Class
Canteen
Office
& Server Room
Teacher's Room
12 IPS Class
12 MA Class
11 IPS Class
11 MA Class
10 IPS Class
10 MA Class

CONFIGURATION ON DEVICE

IP Configuration on DHCP Server



VLAN Configuration on DHCP Server



Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
labPOOL	192.168.40.1	8.8.8.8	192.168.40.3	255.255.255.0	253	0.0.0.0	0.0.0.0
guruPOOL	192.168.30.1	8.8.8.8	192.168.30.3	255.255.255.0	253	0.0.0.0	0.0.0.0
siswaPOOL	192.168.20.1	8.8.8.8	192.168.20.3	255.255.255.0	253	0.0.0.0	0.0.0.0
muridPOOL	192.168.10.1	8.8.8.8	192.168.10.3	255.255.255.0	253	0.0.0.0	0.0.0.0
serverPool	192.168.99.1	8.8.8.8	192.168.99.3	255.255.255.0	253	0.0.0.0	0.0.0.0

CLI COnfiguration on Multilayer Switch (3650-24PS)

CLI Switch>

```
#SET VLAN ON SWITCH FOR NEW AP
enable
configure terminal
vlan 10
name MURID
exit
interface gigabitEthernet1/0/3
switchport mode access
switchport access vlan 10
no shutdown
exit
```

```
#CREATE DHCP FOR NEW VLAN
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
ip dhcp pool MURID_POOL
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8

#SET VLAN ON SWITCH FOR NEW AP
enable
configure terminal
vlan 20
name SISWA
exit
interface gigabitEthernet1/0/6
switchport mode access
switchport access vlan 20
no shutdown
exit
#CREATE DHCP FOR NEW VLAN
interface vlan 20
ip address 192.168.20.1 255.255.255.0
no shutdown
exit
ip dhcp pool SISWA_POOL
network 192.168.20.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

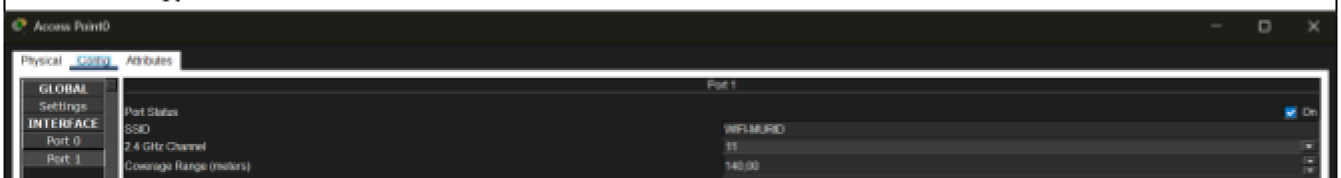
```
#SET VLAN ON SWITCH FOR NEW AP
enable
configure terminal
vlan 40
name LAB
exit
interface gigabitEthernet1/0/4
switchport mode access
switchport access vlan 40
no shutdown
exit
#CREATE DHCP FOR NEW VLAN
interface vlan 40
ip address 192.168.40.1 255.255.255.0
no shutdown
exit
ip dhcp pool LAB_POOL
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 8.8.8.8

#SET VLAN ON SWITCH FOR NEW AP
enable
configure terminal
vlan 30
name GURU
exit
interface gigabitEthernet1/0/5
switchport mode access
switchport access vlan 30
no shutdown
exit
```

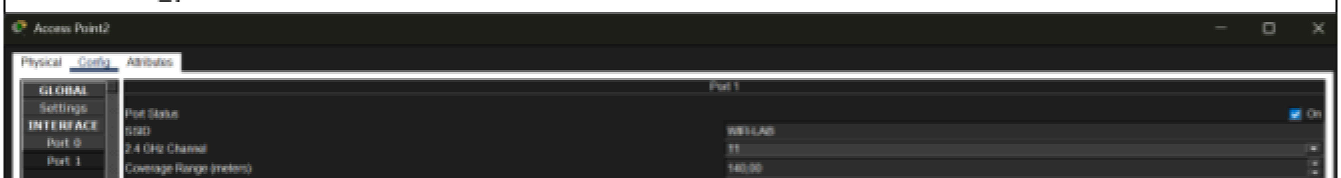
#CREATE DHCP FOR NEW VLAN
interface vlan 30
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
ip dhcp pool GURU_POOL
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
dns-server 8.8.8.8

Configuration on Access Points

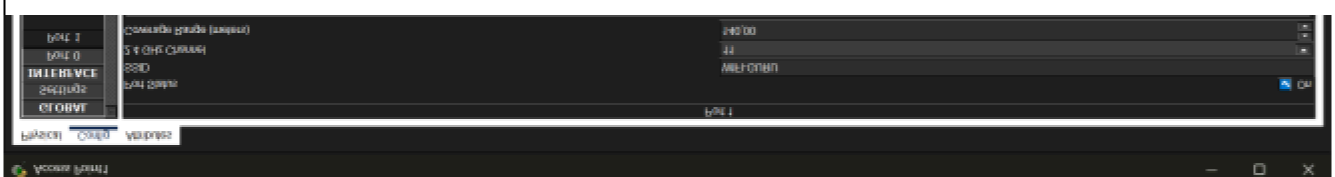
1.



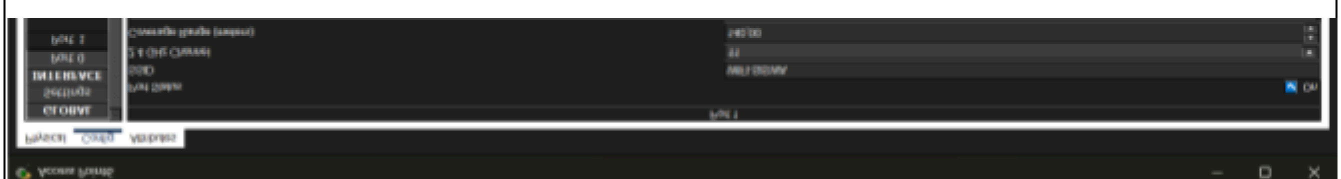
2.



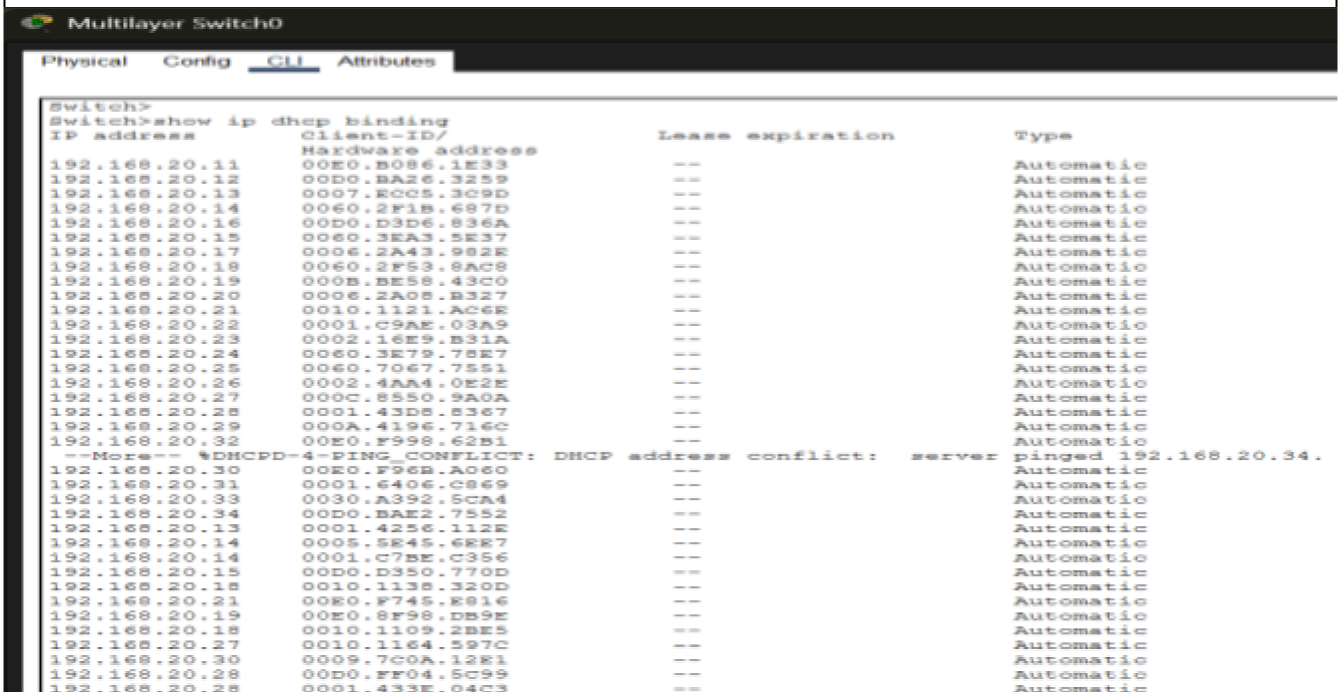
3.



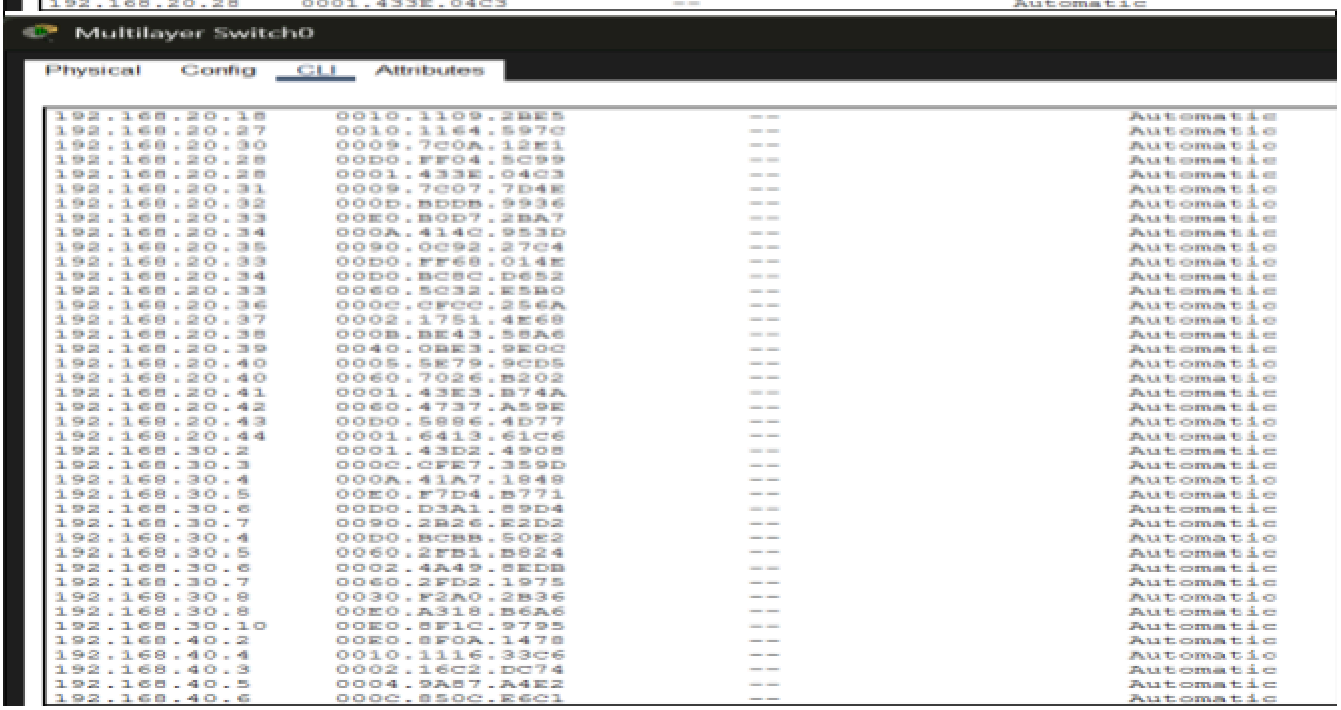
4.



Check if devices is connected in server



```
Switch>
Switch>show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.20.11    00E0.B086.1E33           --                     Automatic
192.168.20.12    00D0.BA26.3259           --                     Automatic
192.168.20.13    0007.ECC5.3C9D           --                     Automatic
192.168.20.14    0060.2F1B.687D           --                     Automatic
192.168.20.16    00D0.D3D6.836A           --                     Automatic
192.168.20.15    0060.3EA3.5E37           --                     Automatic
192.168.20.17    0006.2A43.982E           --                     Automatic
192.168.20.18    0060.2F53.8AC8           --                     Automatic
192.168.20.19    000B.BE58.43C0           --                     Automatic
192.168.20.20    0006.2A08.B327           --                     Automatic
192.168.20.21    0010.1121.AC6E           --                     Automatic
192.168.20.22    0001.C9AE.03A9           --                     Automatic
192.168.20.23    0002.16E9.B31A           --                     Automatic
192.168.20.24    0060.3E79.78E7           --                     Automatic
192.168.20.25    0060.7067.7551           --                     Automatic
192.168.20.26    0002.4AA4.0E2E           --                     Automatic
192.168.20.27    000C.8550.9A0A           --                     Automatic
192.168.20.28    0001.43D8.8367           --                     Automatic
192.168.20.29    000A.4196.716C           --                     Automatic
192.168.20.32    00E0.F998.62B1           --                     Automatic
 --More-- %DHCPD-4-PING_CONFLICT: DHCP address conflict:  server pinged 192.168.20.34.
192.168.20.30    00E0.F96B.A060           --                     Automatic
192.168.20.31    0001.6406.C869           --                     Automatic
192.168.20.33    0030.A392.5CA4           --                     Automatic
192.168.20.34    00D0.BAE2.7552           --                     Automatic
192.168.20.13    0001.4256.112E           --                     Automatic
192.168.20.14    0005.5E45.6EE7           --                     Automatic
192.168.20.14    0001.C7BE.C356           --                     Automatic
192.168.20.15    00D0.D350.770D           --                     Automatic
192.168.20.18    0010.1138.320D           --                     Automatic
192.168.20.21    00E0.F745.E816           --                     Automatic
192.168.20.19    00E0.8F98.DB9E           --                     Automatic
192.168.20.18    0010.1109.2BE5           --                     Automatic
192.168.20.27    0010.1164.597C           --                     Automatic
192.168.20.30    0009.7C0A.12E1           --                     Automatic
192.168.20.28    00D0.FF04.5C99           --                     Automatic
192.168.20.28    0001.433E.04C3           --                     Automatic
```

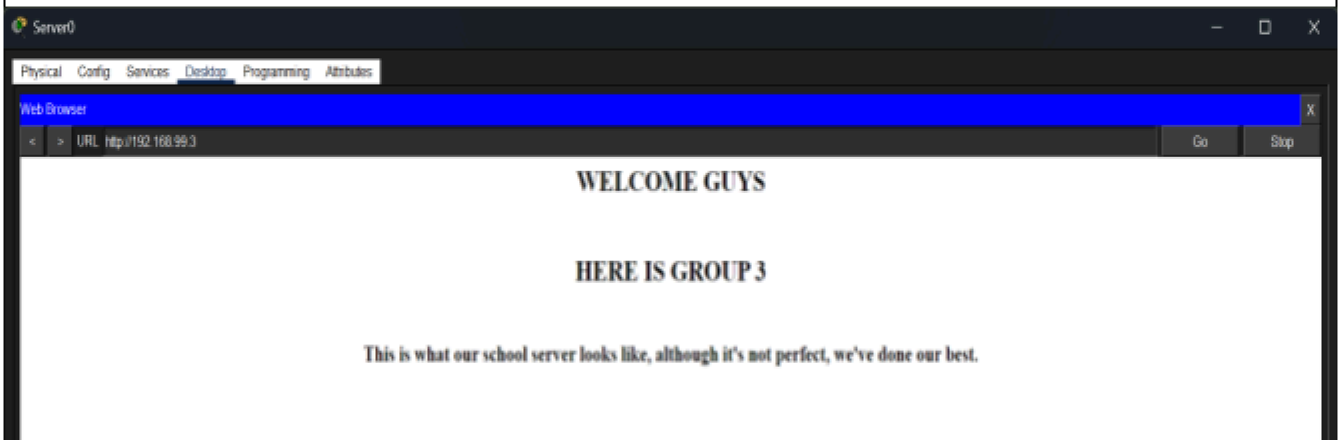


```
192.168.20.18    0010.1109.2BE5           --                     Automatic
192.168.20.27    0010.1164.597C           --                     Automatic
192.168.20.30    0009.7C0A.12E1           --                     Automatic
192.168.20.28    00D0.FF04.5C99           --                     Automatic
192.168.20.28    0001.433E.04C3           --                     Automatic
192.168.20.31    0009.7C07.7D4E           --                     Automatic
192.168.20.32    000D.BDDB.9936           --                     Automatic
192.168.20.33    00E0.B0D7.2BA7           --                     Automatic
192.168.20.34    000A.414C.953D           --                     Automatic
192.168.20.35    0090.0C92.27C4           --                     Automatic
192.168.20.33    00D0.FF68.014E           --                     Automatic
192.168.20.34    00D0.BC8C.D652           --                     Automatic
192.168.20.33    0060.5C32.E5B0           --                     Automatic
192.168.20.36    000C.CFCC.256A           --                     Automatic
192.168.20.37    0002.1751.4E68           --                     Automatic
192.168.20.38    000B.BE43.58A6           --                     Automatic
192.168.20.39    0040.0BE3.9E0C           --                     Automatic
192.168.20.40    0005.5E79.9CD5           --                     Automatic
192.168.20.40    0060.7026.B202           --                     Automatic
192.168.20.41    0001.43E3.B74A           --                     Automatic
192.168.20.42    0060.4737.A59E           --                     Automatic
192.168.20.43    00D0.5886.4D77           --                     Automatic
192.168.20.44    0001.6413.61C6           --                     Automatic
192.168.30.2     0001.43D2.4908           --                     Automatic
192.168.30.3     000C.CFE7.359D           --                     Automatic
192.168.30.4     000A.41A7.1848           --                     Automatic
192.168.30.5     00E0.F7D4.B771           --                     Automatic
192.168.30.6     00D0.D3A1.89D4           --                     Automatic
192.168.30.7     0090.2B26.E2D2           --                     Automatic
192.168.30.4     00D0.BCBB.50E2           --                     Automatic
192.168.30.5     0060.2FB1.B824           --                     Automatic
192.168.30.6     0002.4A49.8EDB           --                     Automatic
192.168.30.7     0060.2FD2.1975           --                     Automatic
192.168.30.8     0030.F2A0.2B36           --                     Automatic
192.168.30.8     00E0.A318.B6A6           --                     Automatic
192.168.30.10    00E0.8F1C.9795           --                     Automatic
192.168.40.2     00E0.8F0A.1478           --                     Automatic
192.168.40.4     0010.1116.33C6           --                     Automatic
192.168.40.3     0002.16C2.DC74           --                     Automatic
192.168.40.5     0004.9A87.A4E2           --                     Automatic
192.168.40.6     000C.850C.E6C1           --                     Automatic
```

```
192.168.40.7      0001.6482.B413     --     Automatic
192.168.40.8      0007.EC90.A55A     --     Automatic
192.168.40.9      000C.CF22.4A8A     --     Automatic
192.168.40.10     00D0.5817.8161     --     Automatic
192.168.40.12     0002.16A1.8DEB     --     Automatic
192.168.40.11     000C.CF97.7692     --     Automatic
192.168.40.13     0001.63E6.B59C     --     Automatic
192.168.40.14     0001.97E3.0C24     --     Automatic
192.168.10.3      00E0.F97C.4A5C     --     Automatic
192.168.10.2      0060.4792.096C     --     Automatic
192.168.10.4      000D.BDB7.4E51     --     Automatic
192.168.10.5      0001.C96C.E6CD     --     Automatic
192.168.10.6      00D0.BC27.AE02     --     Automatic
192.168.10.7      0030.F2C4.7734     --     Automatic
192.168.10.10     0002.1779.AB09     --     Automatic
192.168.10.8      00D0.9731.7CA4     --     Automatic
192.168.10.9      0090.216C.D122     --     Automatic
192.168.10.11     0009.7CB8.D3D1     --     Automatic
192.168.10.13     000A.F3CA.2E13     --     Automatic
192.168.10.12     0001.42C6.B158     --     Automatic
192.168.10.14     0001.429D.69B0     --     Automatic
192.168.10.15     000D.BDC4.3CDB     --     Automatic
192.168.10.5      0002.16BD.71D2     --     Automatic
192.168.10.6      0090.2BA2.DC32     --     Automatic
192.168.10.6      0060.4740.B670     --     Automatic
```

```
192.168.10.6      0060.4740.B670     --     Automatic
192.168.10.8      000D.BD25.52C7     --     Automatic
192.168.10.9      0000.0C34.C438     --     Automatic
192.168.10.10     0006.2A87.51E8     --     Automatic
192.168.10.11     0060.2F46.452B     --     Automatic
192.168.10.12     00D0.D346.CB7C     --     Automatic
192.168.10.13     0030.F230.3B07     --     Automatic
192.168.10.12     0060.479C.4250     --     Automatic
192.168.10.16     0001.C96D.717A     --     Automatic
192.168.10.17     0060.4714.1072     --     Automatic
192.168.10.18     00E0.F79A.3DC2     --     Automatic
Switch>
```

Display in the server browser



SIMULATION

Connection testing to 4 devices in every division on device

```
C:\>ping 192.168.10.18

Pinging 192.168.10.18 with 32 bytes of data:

Reply from 192.168.10.18: bytes=32 time<1ms TTL=128
Reply from 192.168.10.18: bytes=32 time=2ms TTL=128
Reply from 192.168.10.18: bytes=32 time=3ms TTL=128
Reply from 192.168.10.18: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.10.18:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 2ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.40.9

Pinging 192.168.40.9 with 32 bytes of data:

Reply from 192.168.40.9: bytes=32 time<1ms TTL=128
Reply from 192.168.40.9: bytes=32 time<1ms TTL=128
Reply from 192.168.40.9: bytes=32 time=3ms TTL=128
Reply from 192.168.40.9: bytes=32 time=15ms TTL=128

Ping statistics for 192.168.40.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 15ms, Average = 4ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.3

Pinging 192.168.30.3 with 32 bytes of data:

Reply from 192.168.30.3: bytes=32 time<1ms TTL=128
Reply from 192.168.30.3: bytes=32 time=2ms TTL=128
Reply from 192.168.30.3: bytes=32 time=1ms TTL=128
Reply from 192.168.30.3: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.30.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

```
C:\>ping 192.168.20.38

Pinging 192.168.20.38 with 32 bytes of data:

Reply from 192.168.20.38: bytes=32 time=7ms TTL=128
Reply from 192.168.20.38: bytes=32 time=2ms TTL=128
Reply from 192.168.20.38: bytes=32 time=3ms TTL=128
Reply from 192.168.20.38: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.20.38:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 7ms, Average = 3ms

C:\>
```